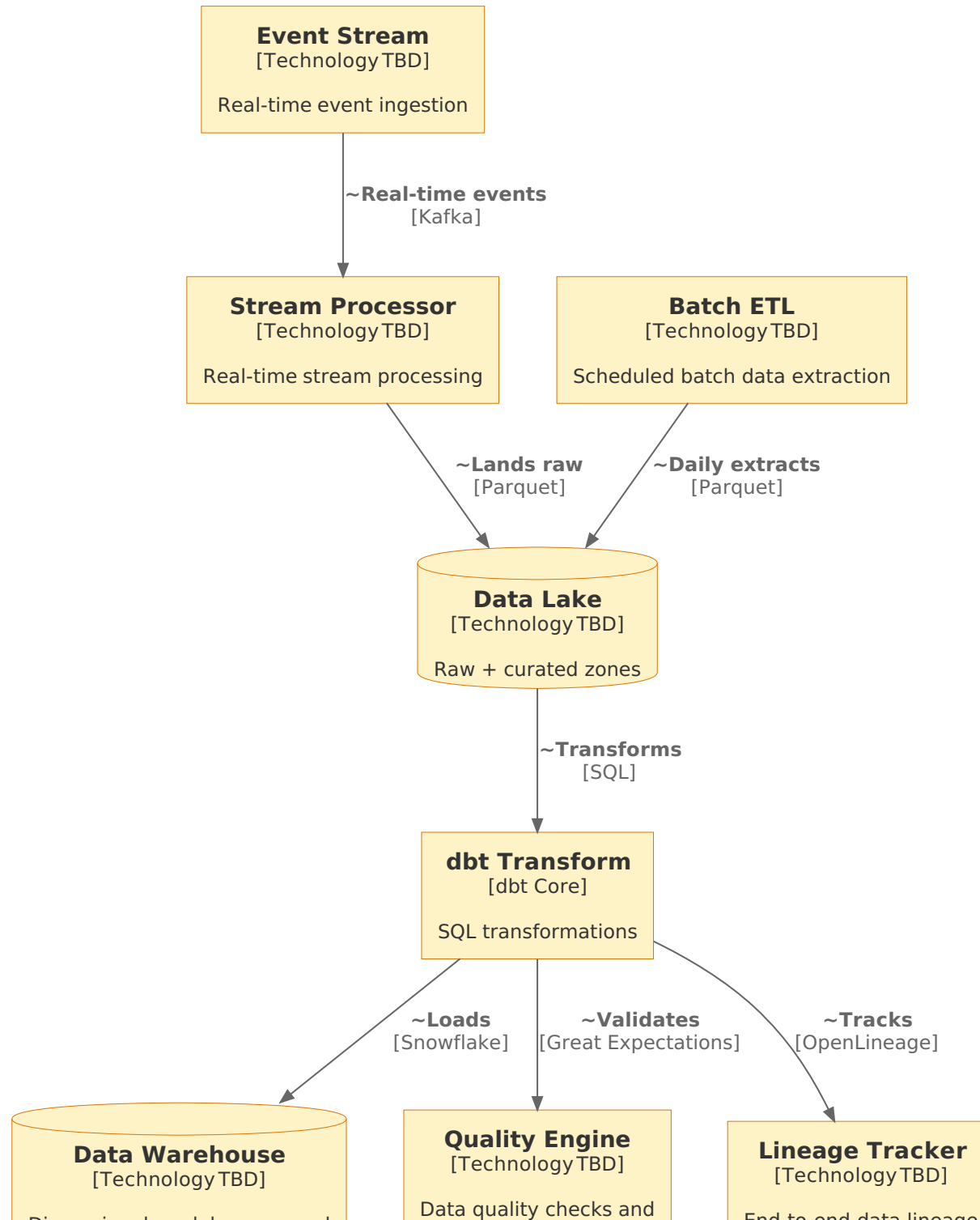


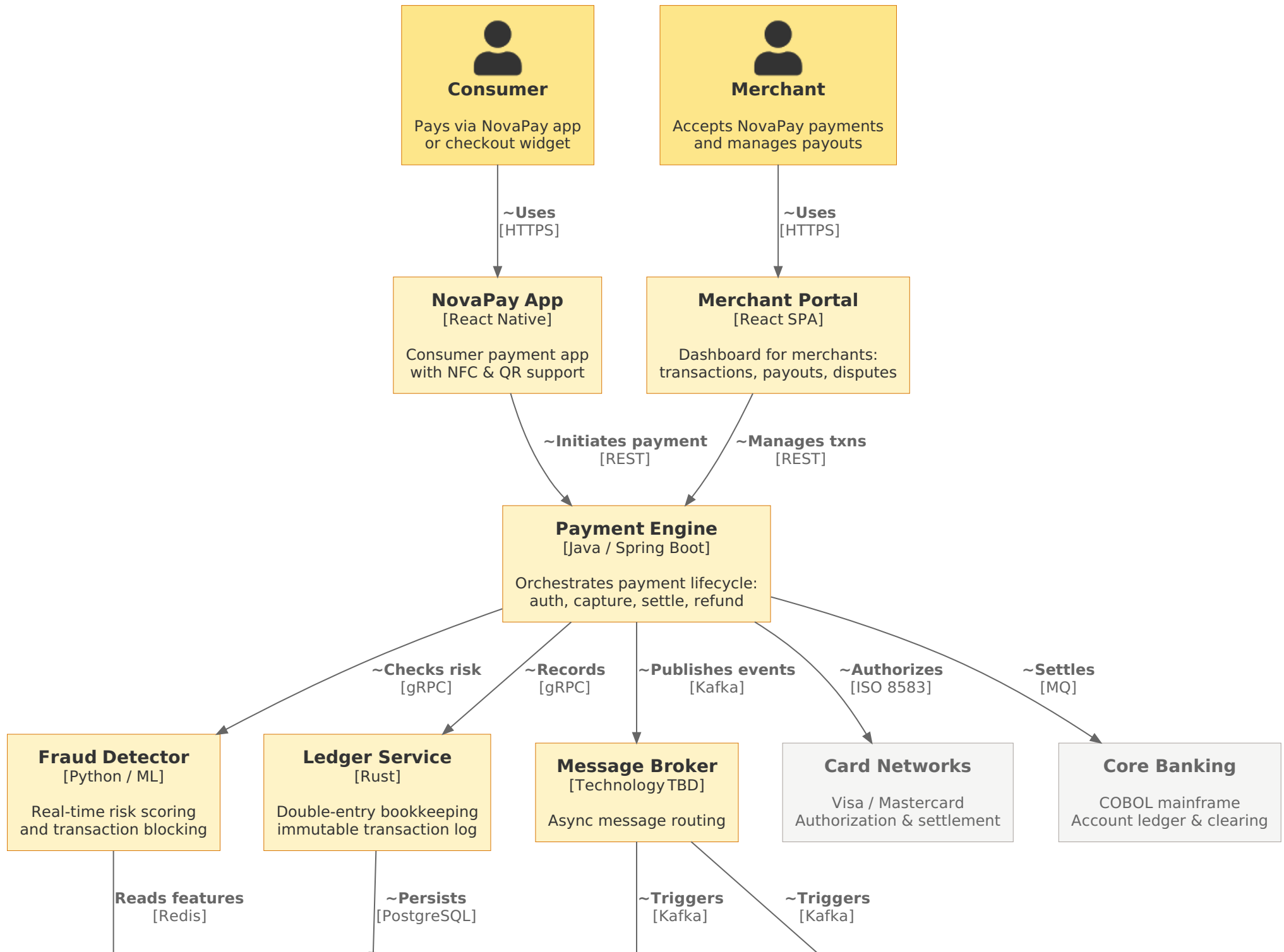
Demo

100%

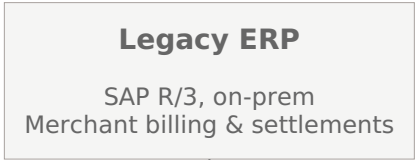
NovaPay — Analytics Platform



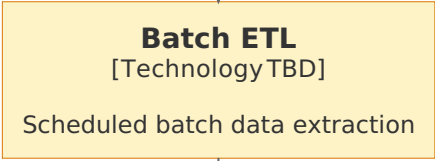
NovaPay — Container Overview



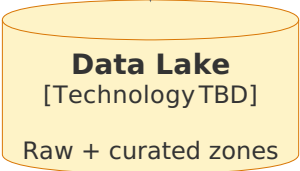
Demo Client — Data Platform



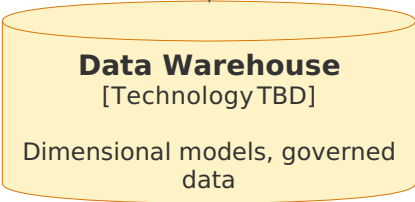
~Exports data
[CSV/FTP]



~Loads to
[Parquet]

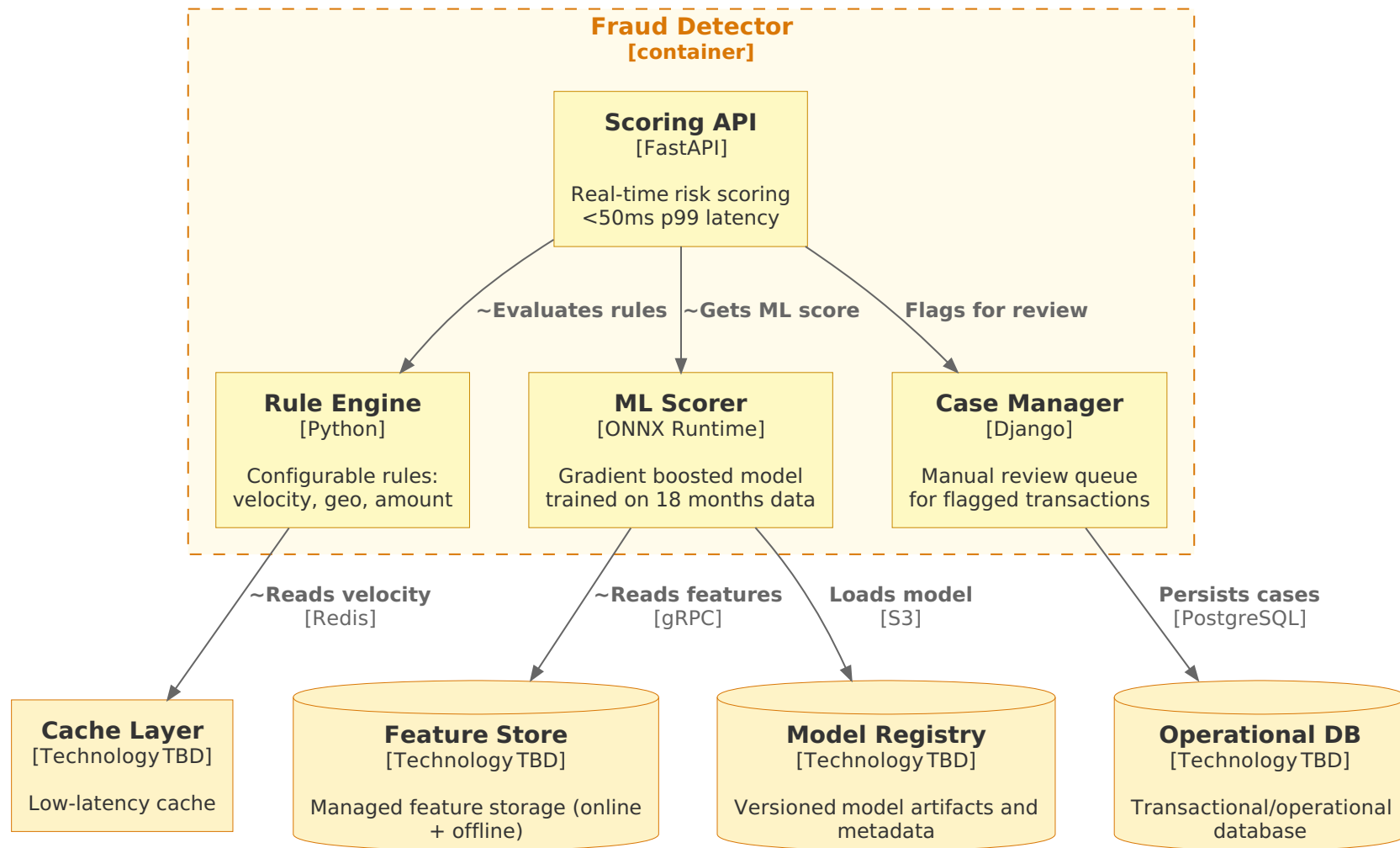


Transforms
[dbt]



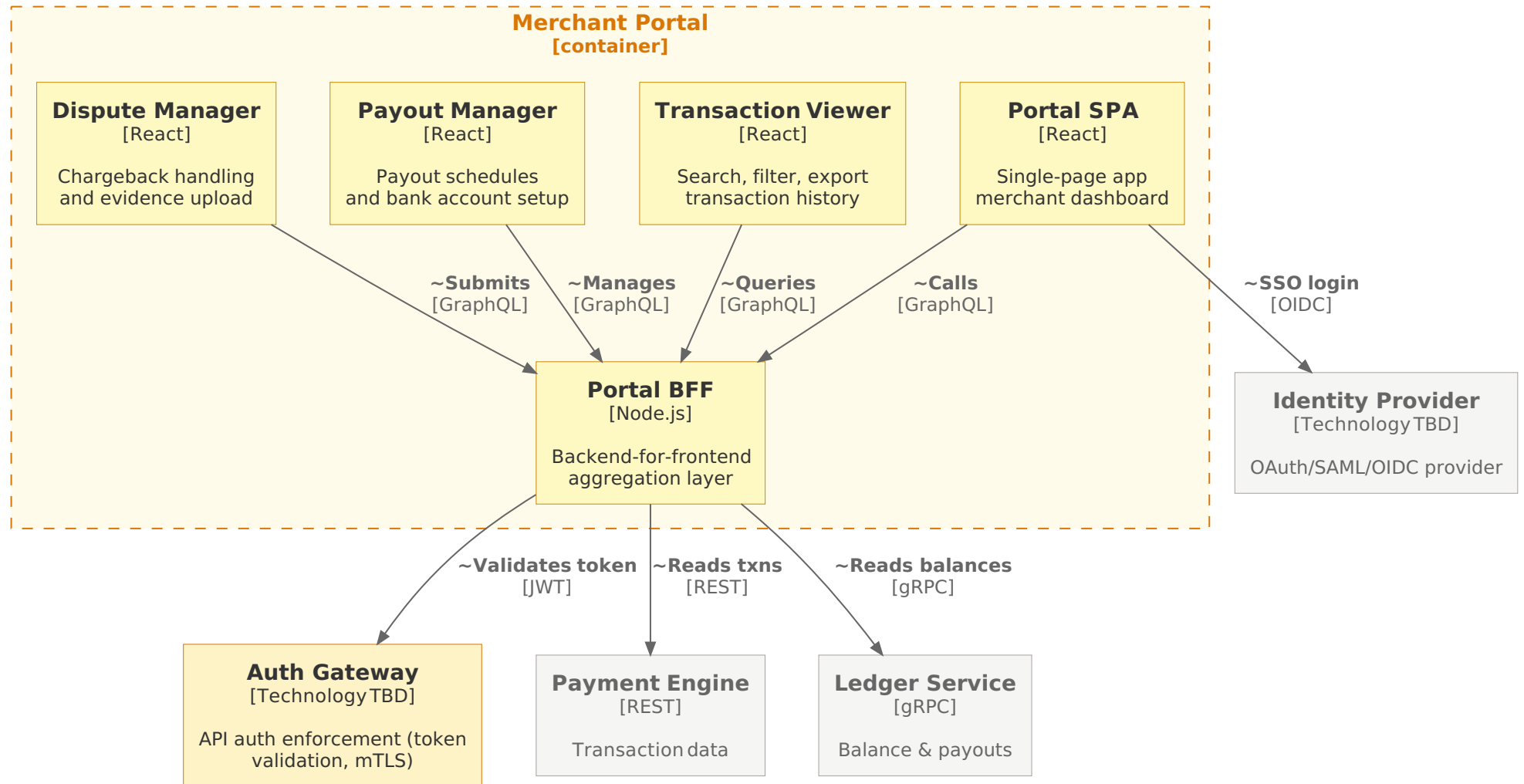
Legend
person
system
container
external person
external system
external container

NovaPay — Fraud Detection System



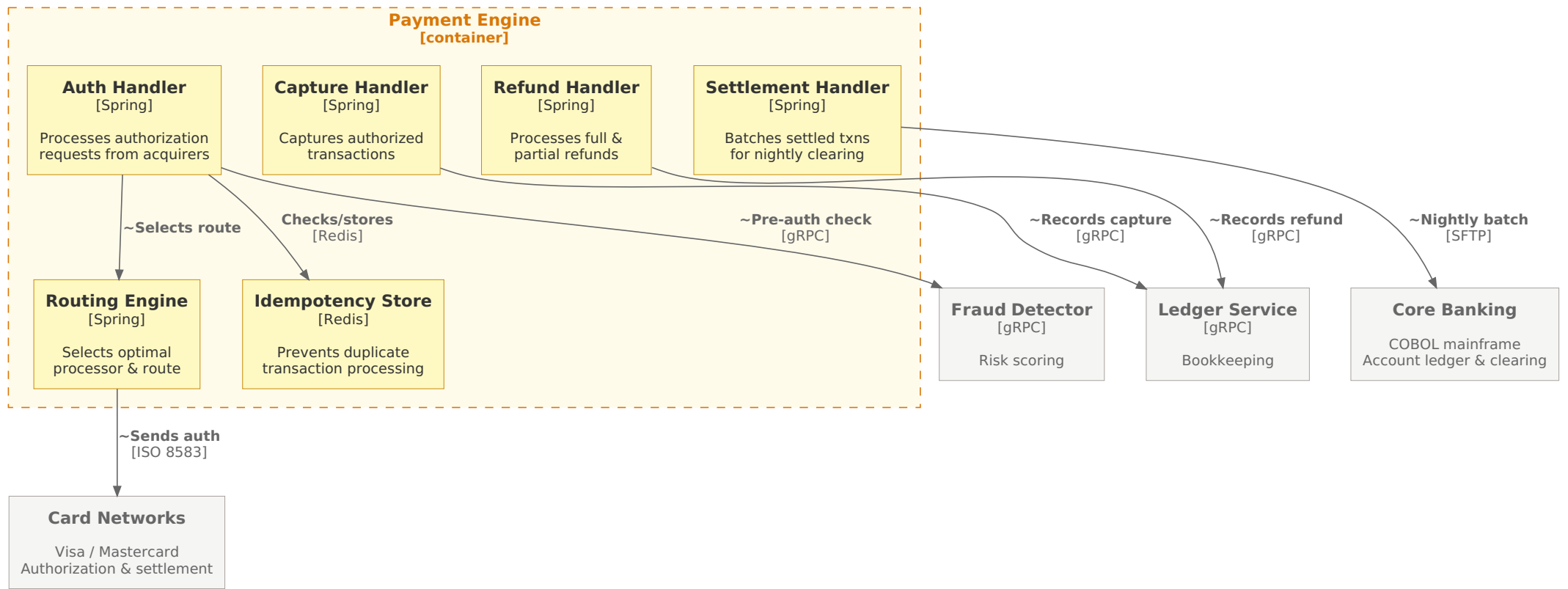
Legend
person
system
container
component
external person
external system
external container
external component

NovaPay — Merchant Portal Components

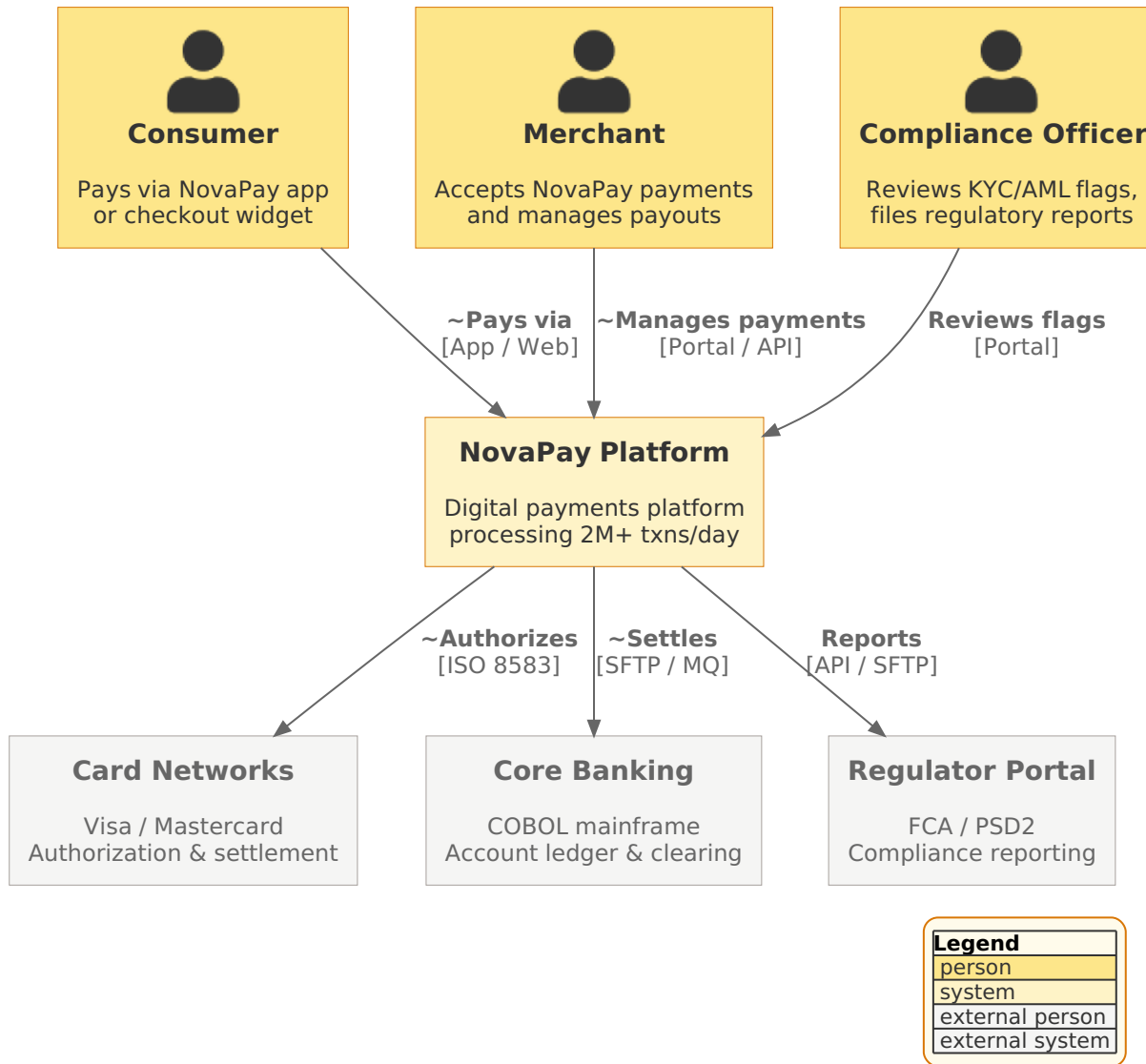


Legend
person
system
container
component
external person
external system
external container
external component

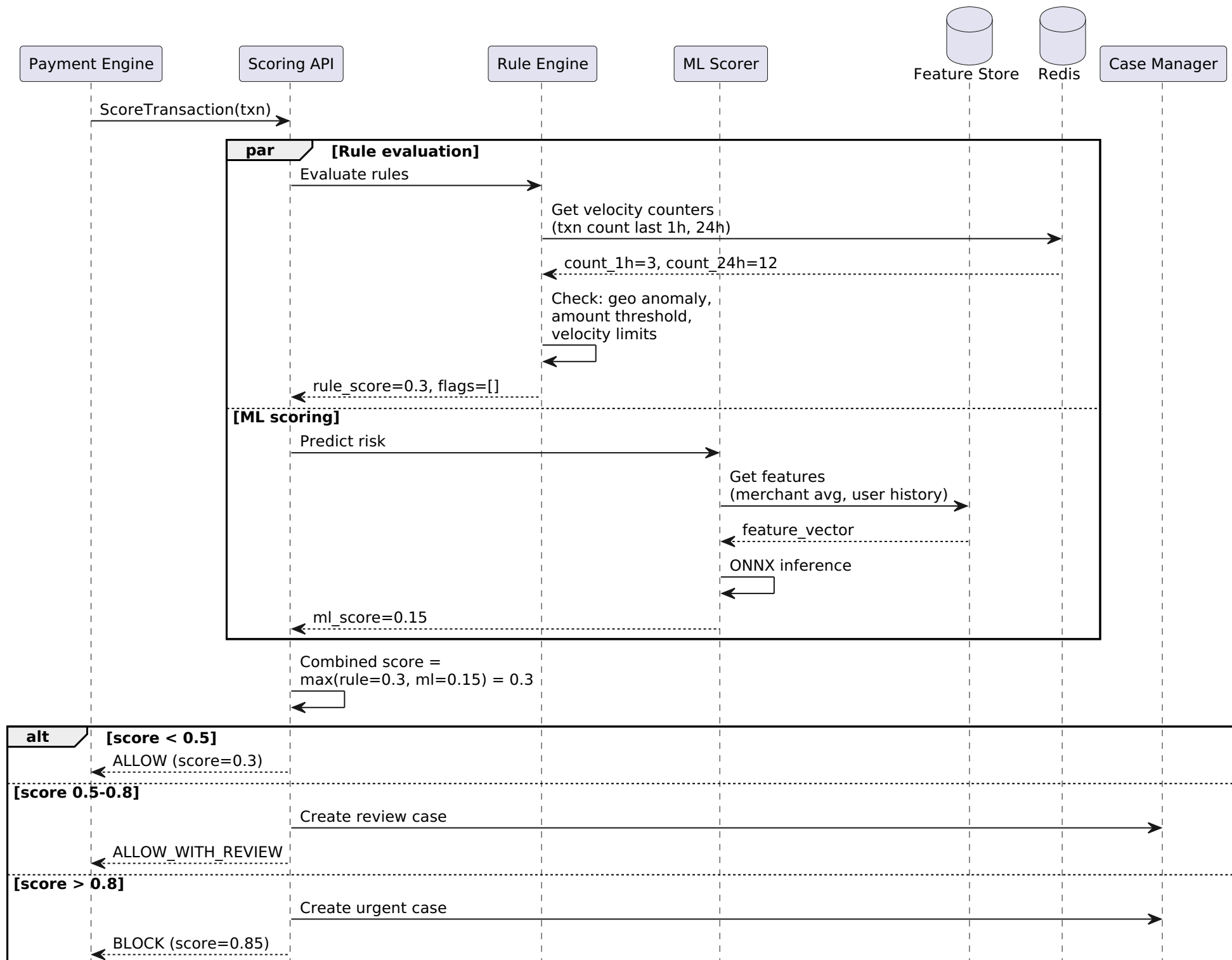
NovaPay — Payment Engine Components



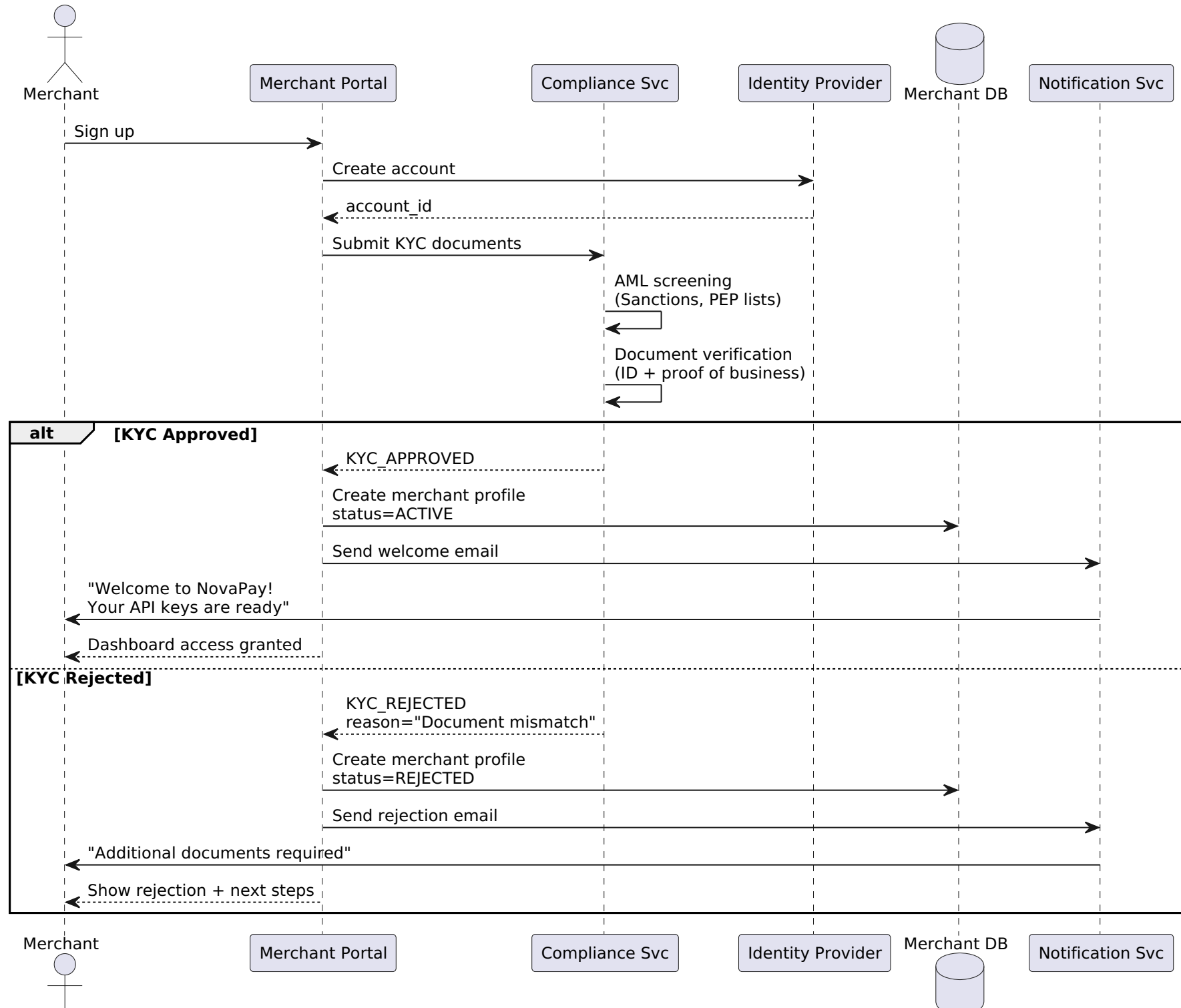
NovaPay — System Context



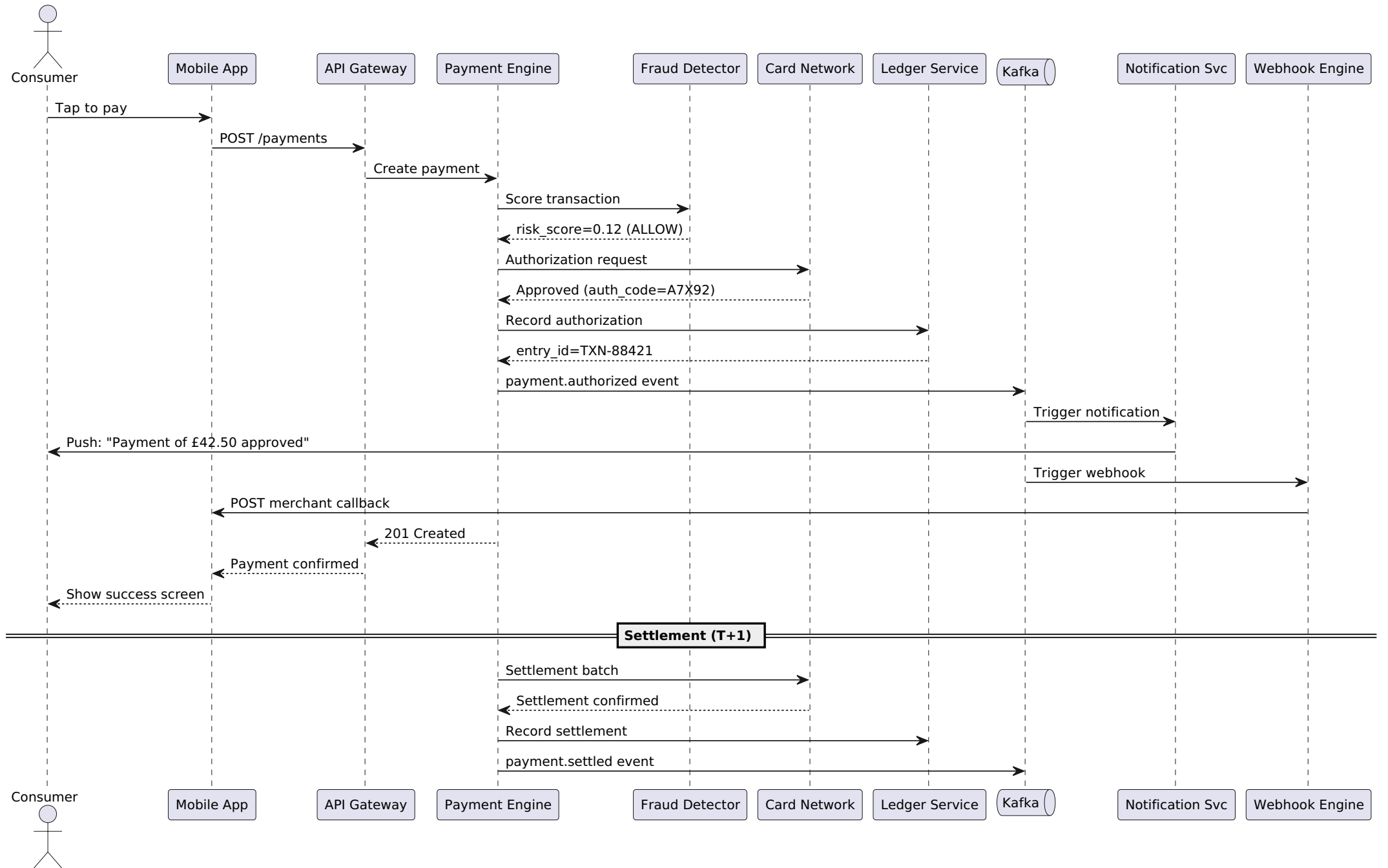
NovaPay — Real-Time Fraud Check



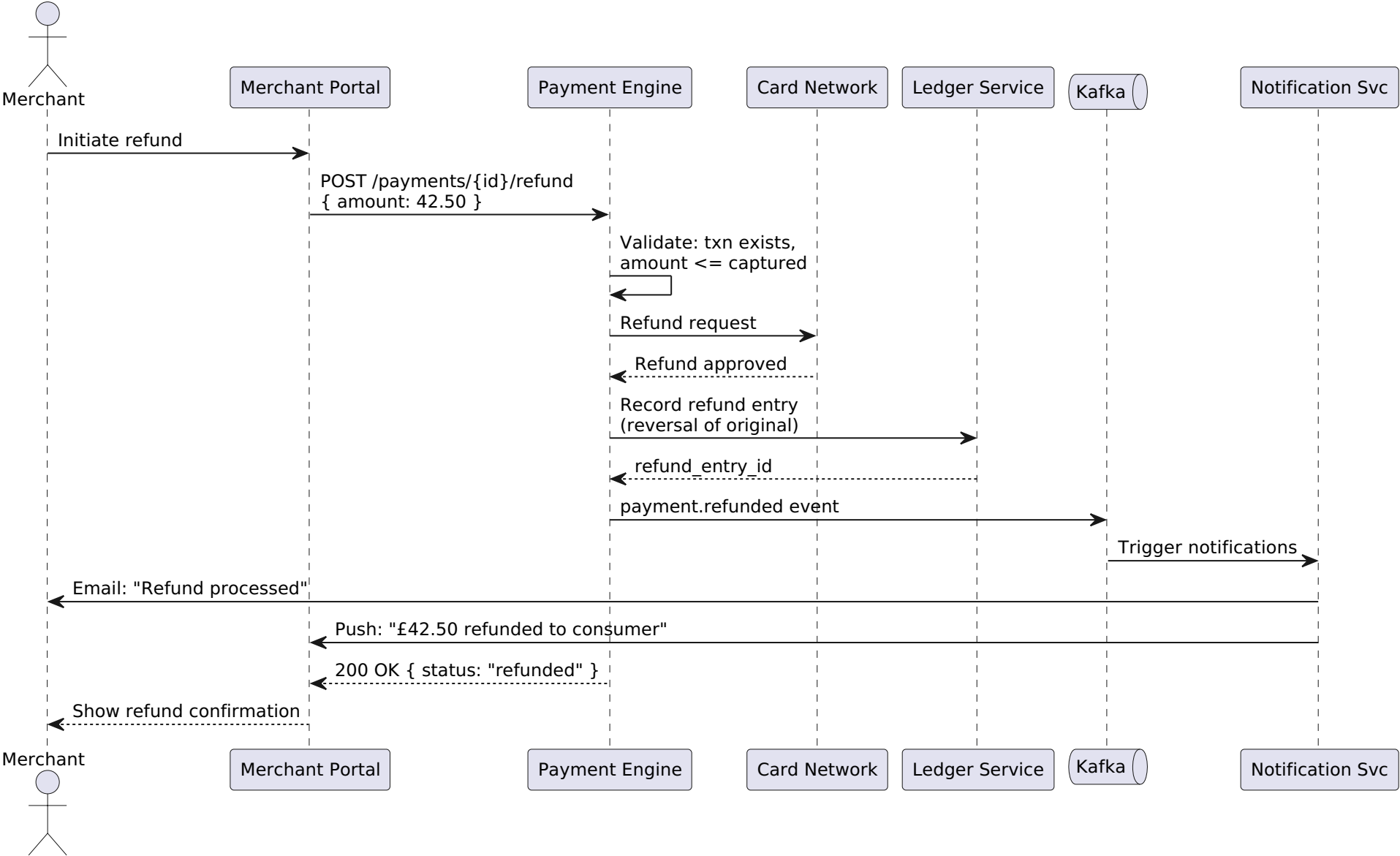
NovaPay — Merchant Onboarding



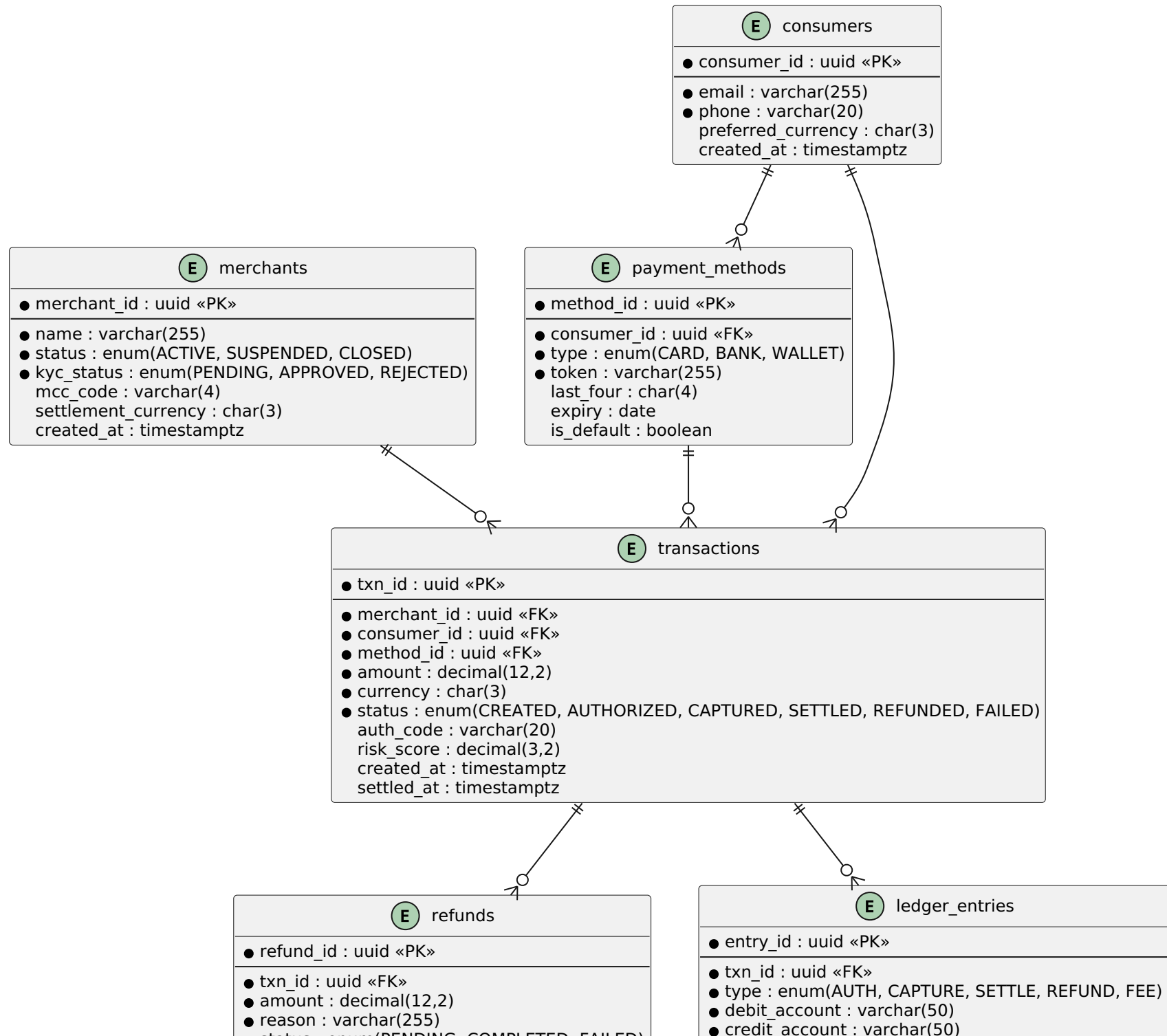
NovaPay — Payment Processing Flow



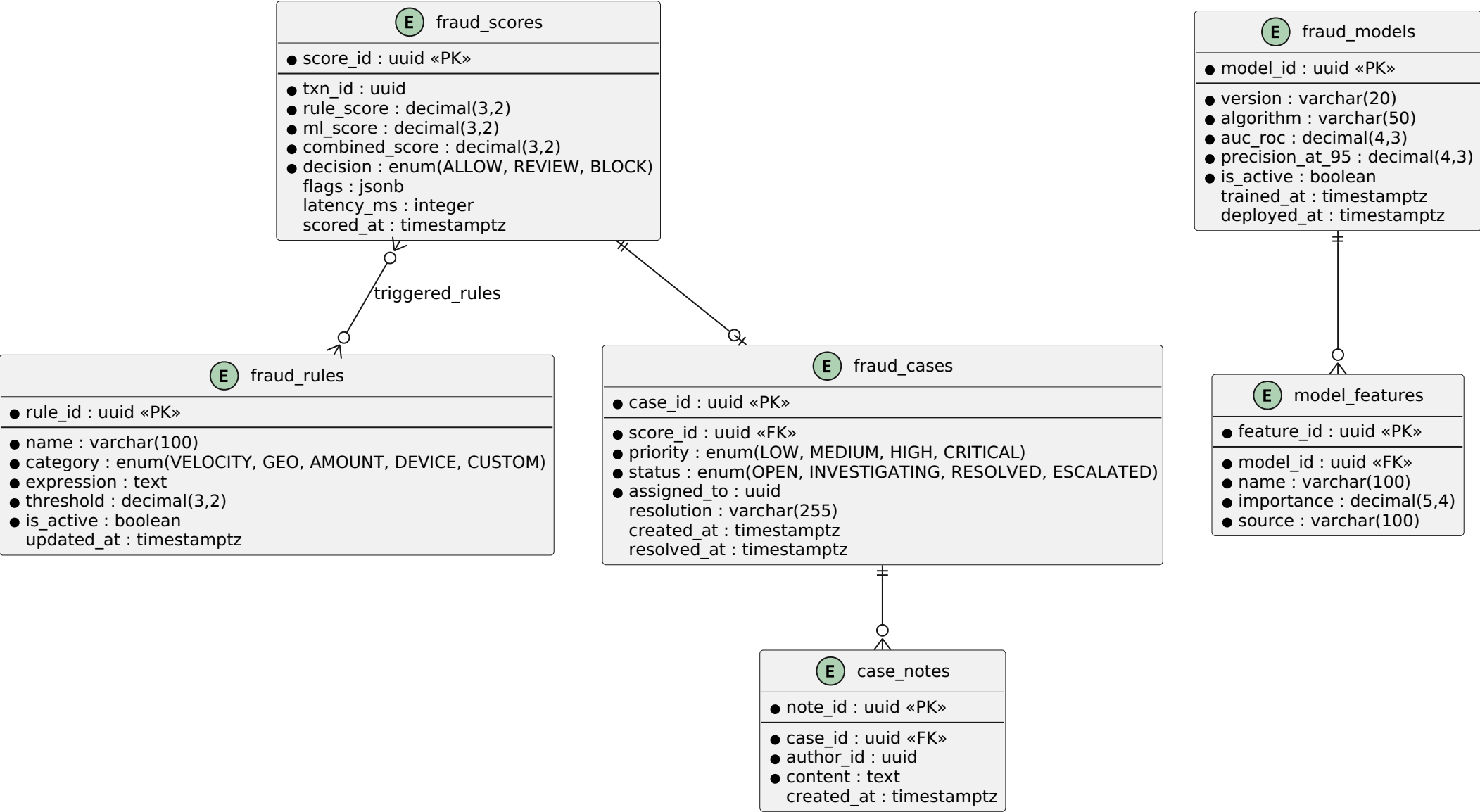
NovaPay — Refund Processing Flow



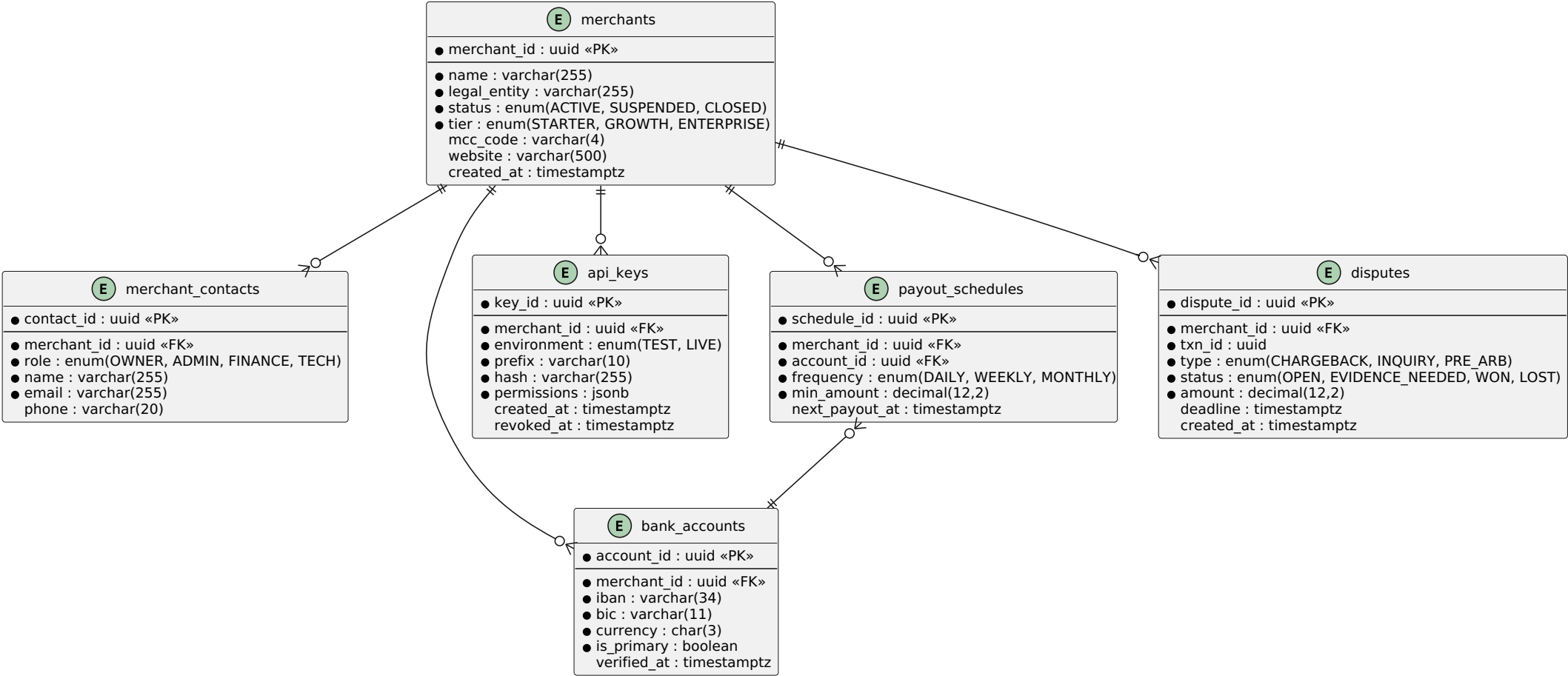
NovaPay — Core Transaction Data Model



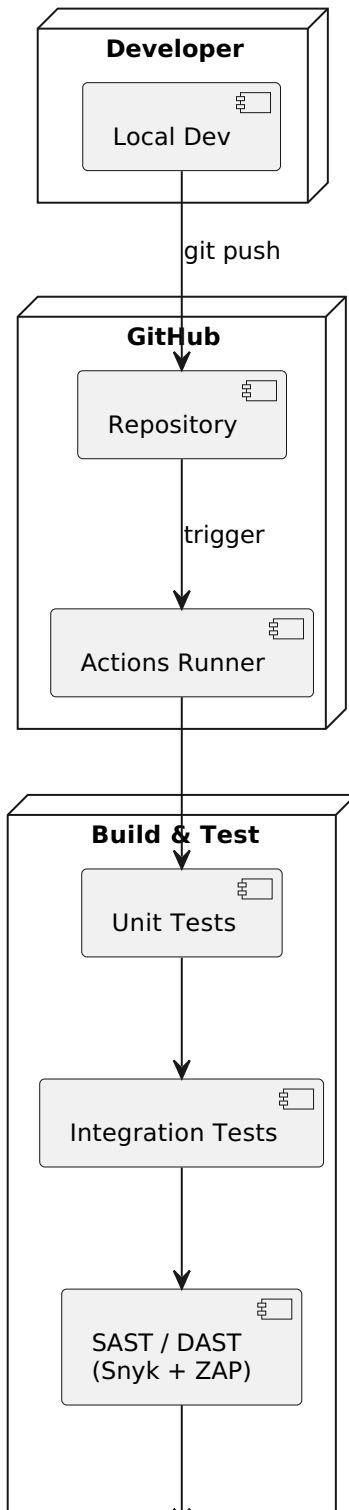
NovaPay — Fraud Case Data Model



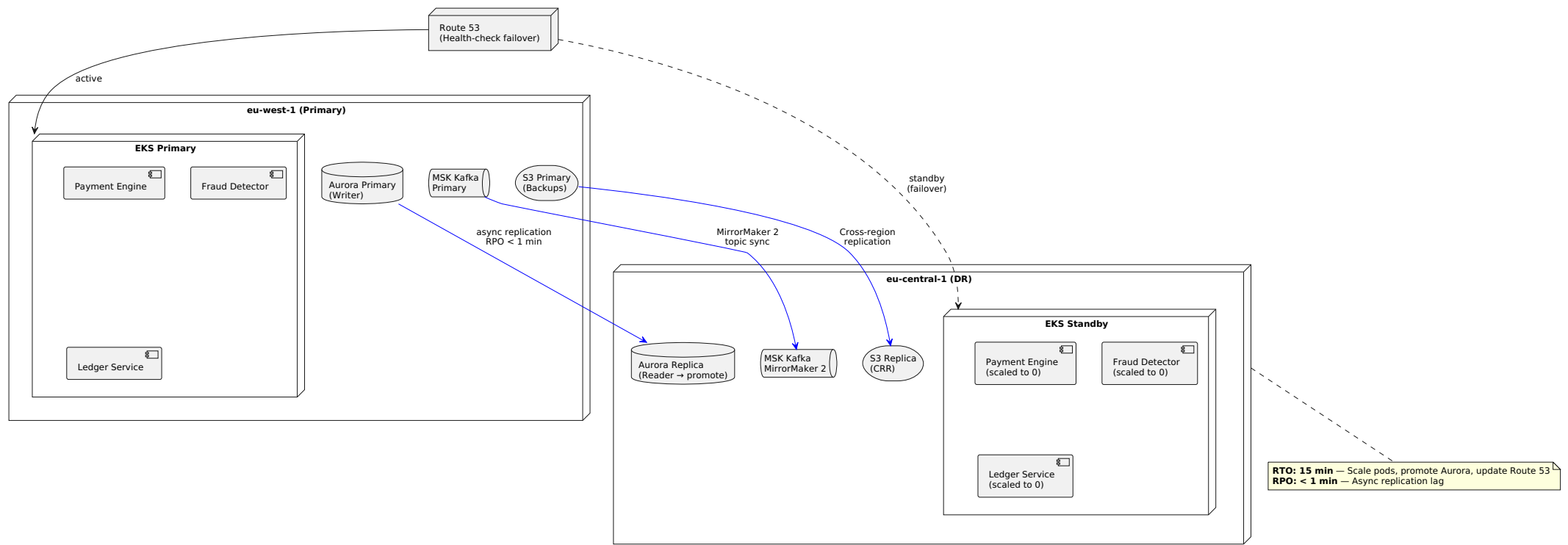
NovaPay — Merchant Management Data Model



NovaPay — CI/CD Pipeline



NovaPay — Disaster Recovery Architecture



NovaPay — Production Infrastructure

AWS eu-west-1 (Primary)

VPC — 10.0.0.0/16

Public Subnet

CloudFront

ALB

Private Subnet — App Tier

EKS Cluster

Payment Engine
(6 pods)

Notification Svc
(2 pods)

Webhook Engine
(2 pods)

Fraud Detector
(4 pods)

Ledger Service
(3 pods)

Private Subnet — Data Tier

ElastiCache Redis
(Cluster mode)

Aurora PostgreSQL
(Multi-AZ)

MSK Kafka
(3 brokers)

External

Kroki API

Visa / Mastercard

AWS eu-central-1 (DR)

EKS Standby

Aurora Read Replica

async replication

